

Intermediate Microeconomics

Week 2 · Day 2 — Welfare and Tax Incidence (Perloff, Chs. 8–9)

Welfare measurement, price controls, and the incidence of a sales tax

Recap of Day 1, and today's plan

Yesterday we solved the market $Q_d = 120 - 2P$, $Q_s = -60 + 4P$:

$$P^* = 30, \quad Q^* = 60, \quad \varepsilon_d = -1, \quad \eta_s = +2.$$

We flagged that *supply is more elastic than demand here* ($\eta_s > |\varepsilon_d|$). Today that single fact decides who bears a tax.

Plan

Consumer & producer surplus	how we <i>measure</i> gains from trade
Total surplus & efficiency	why the competitive outcome is the benchmark
Price ceiling / price floor	binding controls, shortages, surpluses, DWL
Sales tax	incidence, pass-through, revenue, deadweight loss

The same (30, 60) equilibrium anchors every intervention today.

Learning goals — Day 2

- ① Interpret the demand curve as marginal willingness to pay, and supply as marginal cost.
- ② Compute consumer, producer, and total surplus for linear curves, and explain why equilibrium is efficient.
- ③ Analyse binding price ceilings and floors: quantity traded, shortage/surplus, deadweight loss, and the distribution of gains and losses.
- ④ Solve for post-tax prices and quantity, whether the tax is levied on buyers or on sellers.
- ⑤ Decompose tax incidence: statutory vs economic, the pass-through fraction, and the elasticity rule.
- ⑥ Separate the transfer (revenue) from the waste (deadweight loss).

Consumer surplus — the idea before the formula

Every buyer has a *willingness to pay* (WTP): the most they would give up rather than go without. The market charges everyone the *same* price P^* . A buyer whose WTP exceeds P^* pockets the difference — that private gain is their **consumer surplus**.

A four-buyer warm-up (price = 30)

Buyer	A	B	C	D	
WTP	55	45	35	25	
Buys at 30?	yes	yes	yes	no	
Surplus	25	15	5	0	$\Rightarrow CS = 45$

Buyer D values the good below its price and walks away — correctly, since forcing that trade would destroy value. Stack many buyers and the WTP “steps” smooth into the **demand curve**: its height at each Q is the marginal buyer’s WTP.

Consumer surplus — computing the area

With a continuum of buyers, CS is the area between the demand curve and the price line, out to Q^* :

$$CS = \int_0^{Q^*} [P_d(Q) - P^*] dQ.$$

Step 1 — inverse demand

$P_d(Q) = 60 - \frac{1}{2}Q$, so the choke (max WTP) is 60.

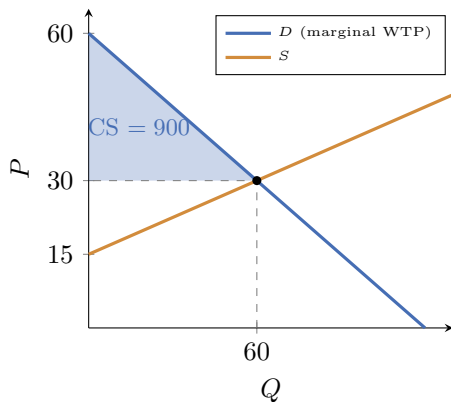
Step 2 — height and base

height = $60 - 30 = 30$, base = $Q^* = 60$.

Step 3 — triangle

$$CS = \frac{1}{2}(60)(30) = \boxed{900}.$$

Numeric check: $\int_0^{60} (60 - \frac{1}{2}Q - 30) dQ = [30Q - \frac{1}{4}Q^2]_0^{60} = 900.$



Producer surplus — the idea before the formula

The mirror image. Each unit has a *marginal cost*: the least a seller would accept rather than not produce it. A seller who receives P^* above that cost earns **producer surplus**.

A four-unit warm-up (price = 30)

Unit	1	2	3	4	
Marginal cost	20	25	30	35	
Produced at 30?	yes	yes	yes	no	
Surplus	10	5	0	0	$\Rightarrow PS = 15$

Unit 4 costs more than the price fetches, so it is left unmade. Stack many units and the cost “steps” smooth into the **supply curve**: its height at each Q is the marginal cost of that unit. (This is why Day 3 can rebuild supply from firms’ MC curves.)

Producer surplus — computing the area

PS is the area between the price line and the supply curve, out to Q^* :

$$PS = \int_0^{Q^*} [P^* - P_s(Q)] dQ.$$

Step 1 — inverse supply

$P_s(Q) = 15 + \frac{1}{4}Q$, so the choke (min acceptable price) is 15.

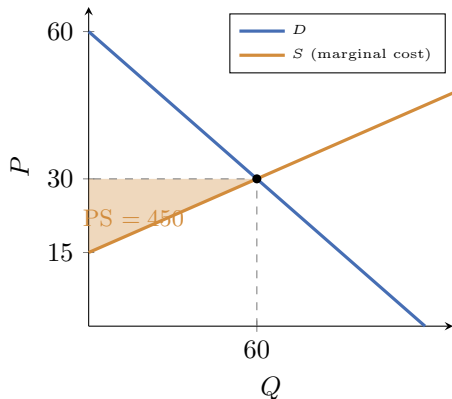
Step 2 — height and base

height = $30 - 15 = 15$, base = $Q^* = 60$.

Step 3 — triangle

$$PS = \frac{1}{2}(60)(15) = \boxed{450}.$$

Numeric check: $\int_0^{60} (30 - 15 - \frac{1}{4}Q) dQ = [15Q - \frac{1}{8}Q^2]_0^{60} = 450$.



Total surplus and why equilibrium is efficient

Total surplus is the whole gains-from-trade triangle:

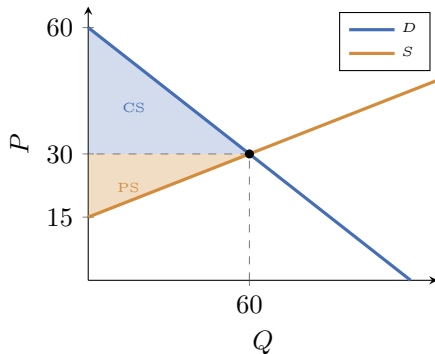
$$TS = CS + PS = 900 + 450 = \boxed{1350}.$$

The efficiency argument

A trade adds value whenever $WTP > MC$:

- $Q = 50$: $WTP\ 35 > MC\ 27.5$ — a *missed* gain.
- $Q = 60$: $WTP = MC = 30$ — nothing left.
- $Q = 70$: $WTP\ 25 < MC\ 32.5$ — a *wasteful* unit.

At $Q^* = 60$ every valuable trade is made and no wasteful one is forced: **equilibrium maximises total surplus**.
Every policy below is a shortfall from 1350.



Price ceilings — motivation

A **price ceiling** is a legal maximum, set below the market price to keep something “affordable.”

Where you have seen them

- **Rent control** — New York City apartments; Berlin’s 2020 rent cap (later struck down).
- **Petrol price controls** — the 1970s US caps that produced hours-long queues at the pump.
- **Anti-price-gouging laws** — caps on water, fuel, generators after hurricanes.
- **Usury laws** — ceilings on interest rates (a price cap on borrowing).

The political logic is obvious — protect buyers from a high price. The economic question is what a binding cap does to *quantity*, and to *whom*. Intuition says “consumers win.” Hold that thought.

A binding ceiling ($\bar{P} = 24 < P^*$): the mechanics

Below P^* , sellers are the *short side* — they decide how much trades.

Step 1 — quantities at $\bar{P} = 24$

$$Q_s = -60 + 4(24) = 36, \quad Q_d = 120 - 48 = 72.$$

Step 2 — shortage and trade

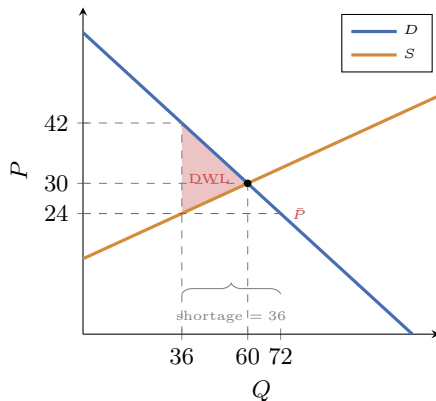
Traded = min = 36; **shortage** = $72 - 36 = 36$.

Step 3 — deadweight loss

The marginal unit ($Q = 36$) is valued at $P_d = 60 - 18 = 42$ but costs only 24; the lost trades run $36 \rightarrow 60$:

$$DWL = \frac{1}{2}(60 - 36)(42 - 24) = \boxed{216}.$$

Numeric check: TS falls $1350 \rightarrow 1134$; loss = 216.



“Consumers win” — do they? The distribution under a ceiling

The tempting conclusion is that a lower price must help consumers. That misses two things: *fewer units exist to buy*, and *someone must be turned away*. So the honest question is **which** consumers, and **how** the scarce units are allocated.

Consumers split in two

- **The lucky** 36 who obtain the good pay 24 instead of 30 — a clear gain.
- **The rationed-out** would-be buyers (up to 72) get nothing; their surplus vanishes.

Net consumer surplus therefore depends on *rationing*: under **efficient** rationing (highest-WTP buyers served) $CS = 972 > 900$; under **random** rationing it is lower and can fall below 900. Producers, by contrast, lose unambiguously: $PS : 450 \rightarrow 162$. And the 216 DWL understates the true loss once queueing and search time are counted.

Price floors — motivation

A **price floor** is a legal minimum, set above the market price to protect *sellers'* incomes.

Where you have seen them

- **The minimum wage** — a floor in the labour market; the “surplus” is unemployment.
- **Agricultural price supports** — the US and the EU’s Common Agricultural Policy, generating butter mountains and grain stores.
- **Minimum unit pricing for alcohol** — Scotland (2018), a floor aimed at public health.

The logic mirrors the ceiling: help the protected side by propping up the price. And the unintended consequence mirrors it too — not a shortage now, but a *glut* the short side won’t absorb.

A binding floor ($\underline{P} = 36 > P^*$): the mechanics

Above P^* , buyers are now the *short side*.

Step 1 — quantities at $\underline{P} = 36$

$$Q_d = 120 - 72 = 48, \quad Q_s = -60 + 144 = 84.$$

Step 2 — glut and trade

Traded = min = 48; **excess supply** = $84 - 48 = 36$.

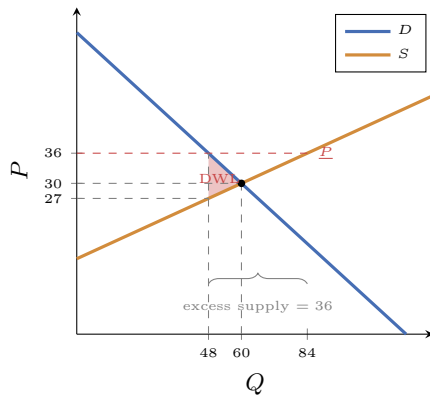
Step 3 — deadweight loss

At $Q = 48$: buyers pay 36; marginal cost = $15 + 12 = 27$.

Lost trades run $48 \rightarrow 60$:

$$DWL = \frac{1}{2}(60 - 48)(36 - 27) = \boxed{54}.$$

Numeric check: Triangle only. If all 84 are produced and 36 dumped, PS falls further and the true loss is larger.



Who bears the floor, and the overproduction trap

Distribution

- **Sellers who sell** the 48 units gain: higher price (36 vs 30).
- **Sellers who cannot sell** the extra 36 units of willing supply are shut out — in a labour market, these are the *unemployed* at the minimum wage.
- **Buyers** lose: higher price, fewer units. $CS : 900 \rightarrow 576$.

The triangle DWL of 54 assumes production stops at the 48 that sell. In practice a floor invites **overproduction**: if producers make all 84 (chasing the high price) and the unsold 36 are stored or destroyed, their cost is pure waste and the loss balloons well past 54 — the economic case against unconditional farm price supports.

Why tax a market, and what a specific tax is

Governments tax transactions to **raise revenue**, to **discourage** consumption (tobacco, alcohol, carbon), or to **correct** an externality (a preview of Pigouvian taxes).

Two forms

- **Specific (per-unit) tax** t : a fixed \$ amount per unit — e.g. \$9 per unit. Today's focus.
- **Ad valorem tax**: a percentage of price — e.g. VAT at 20%. Same logic, scaled by price.

Whatever its purpose, a per-unit tax drives a **wedge** between the two prices in the market:

$$\underbrace{P_d}_{\text{buyers pay}} = \underbrace{P_s}_{\text{sellers keep}} + t.$$

The rest of the section is about who ends up on the wrong side of that wedge.

Levied on buyers or on sellers? Set up both

Statutory incidence is who legally hands the money to the government. Two designs for the *same* \$9 tax:

Tax on buyers

Buyers pay the seller's price *plus* t to the state.
At any seller price, the amount buyers are willing to offer sellers falls by t : the demand curve sellers face **shifts down by t** .

Tax on sellers

Sellers remit t per unit, so they need t more to supply any given quantity: the supply curve buyers face **shifts up by t** .

These look like different policies. The natural guess is that the side *legally charged* is the side that pays. We now solve both and test that guess — watch the traded quantity and the two prices.

Tax on the buyer ($t = 9$): demand shifts down

Buyers pay P_d ; sellers receive $P_s = P_d - 9$. The market clears on quantity.

Step 1 — clearing condition

$$\underbrace{120 - 2P_d}_{Q_d \text{ at buyer price}} = \underbrace{-60 + 4(P_d - 9)}_{Q_s \text{ at seller price}}$$

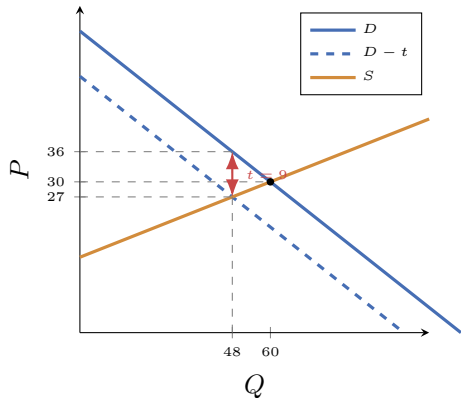
Step 2 — solve

$$120 - 2P_d = -96 + 4P_d \Rightarrow 216 = 6P_d \Rightarrow P_d = 36.$$

Step 3 — read off

$$P_s = 27, \quad Q_t = 120 - 72 = 48.$$

Numeric check: Buyers pay 36 (was 30), sellers net 27, wedge = 9. ✓



The same tax on the seller: supply shifts up — identical outcome

Now sellers remit the \$9. They supply on $Q_s = -60 + 4(P - 9)$ while buyers pay P .

Step 1 — clearing condition

$$120 - 2P = -60 + 4(P - 9)$$

Step 2 — solve

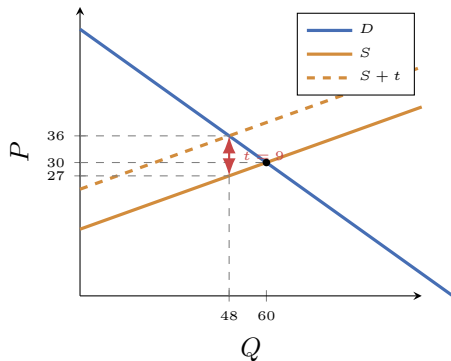
$$120 - 2P = -96 + 4P \Rightarrow 216 = 6P \Rightarrow P = 36.$$

Step 3 — read off

Buyers pay $P_d = 36$; sellers net $36 - 9 = 27$; $Q_t = 48$.

Statutory invariance

Identical to the buyer-side tax — same prices, same quantity. Who legally remits changes **nothing real**.



Incidence: the pass-through fraction

Incidence asks how the \$9 wedge is *split* — how far buyers' price rises versus how far sellers' price falls. Track each from the old $P^* = 30$:

$$P_d : 30 \rightarrow 36 \Rightarrow +6, \quad P_s : 30 \rightarrow 27 \Rightarrow -3.$$

Pass-through to buyers

Buyers absorb \$6 of the \$9: **two-thirds**. Sellers absorb \$3: **one-third**.

Where does 2/3 come from? Differentiate the taxed equilibrium $120 - 2P_d = -60 + 4(P_d - t)$ in t :

$$-2 \frac{dP_d}{dt} = 4 \frac{dP_d}{dt} - 4 \implies \frac{dP_d}{dt} = \frac{4}{6} = \frac{2}{3}.$$

The buyers' share is $\frac{|dQ_s/dP|}{|dQ_s/dP| + |dQ_d/dP|} = \frac{4}{4+2} = \frac{2}{3}$: it is a race between the two slopes.

Incidence in elasticities: the inelastic side pays

Slopes become intuition once written as elasticities at the initial equilibrium ($\varepsilon_d = -1$, $\eta_s = +2$):

$$\text{buyers' share} = \frac{\eta_s}{\eta_s + |\varepsilon_d|}, \quad \text{sellors' share} = \frac{|\varepsilon_d|}{\eta_s + |\varepsilon_d|}$$

$$\text{here} = \frac{2}{2+1} = \frac{2}{3} : \frac{1}{3}.$$

The rule, in words

The **relatively inelastic side bears the larger share**. Being inelastic means you cannot walk away — so you cannot dodge the tax.

Here demand ($|\varepsilon_d| = 1$) is less elastic than supply ($\eta_s = 2$), so consumers carry two-thirds. *Limiting cases*: perfectly inelastic demand ($\varepsilon_d = 0$) $\Rightarrow$ buyers bear all of it; perfectly elastic demand $\Rightarrow$ sellers bear all of it. This is why cigarette taxes fall mostly on smokers.

Tax revenue — a transfer

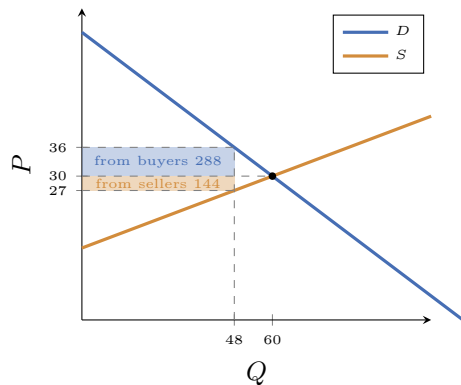
Revenue is the wedge times the traded quantity:

$$R = t \cdot Q_t = 9 \times 48 = \boxed{432}.$$

Split it by incidence:

$$\underbrace{6 \times 48}_{\text{from buyers}} = 288, \quad \underbrace{3 \times 48}_{\text{from sellers}} = 144.$$

This is a **transfer**, not a loss: every dollar leaves buyers and sellers but arrives at the government. On the diagram it is the rectangle between P_d and P_s over the 48 units traded.



Deadweight loss — the waste

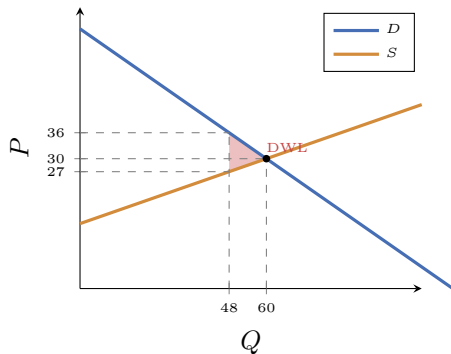
The tax kills the 12 trades between $Q_t = 48$ and $Q^* = 60$ that were worth more to buyers than they cost to make. Their lost value is the red triangle:

$$DWL = \frac{1}{2} t (Q^* - Q_t) = \frac{1}{2} (9)(12) = \boxed{54}.$$

Accounting check

	pre	post	Δ
CS	900	576	-324
PS	450	288	-162
Revenue	0	432	+432
Total	1350	1296	-54

Private surplus falls 486; the state recovers 432; the unrecovered 54 is pure waste. ✓



In-class exercises — Day 2

Time for the in-class exercises.

- **Exercise 1** — welfare and a price ceiling: compute CS , PS , then the shortage and deadweight loss of a binding control.
- **Exercise 2** — tax incidence: split a specific tax using elasticities, show the statute is irrelevant, and find revenue and DWL.

Solutions are at the end of the exercise sheet — try both fully before turning there.